

LOUO Gesture-Recognition Experiment Report

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Configuration

LOUO users	11_Mansh, 12_Marcus, 14_StephenV2, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin
Sensor columns	176 channels
Resample steps	90
Butterworth	True, cutoff=10.0Hz, order=4
Normalisation	minmax
Per-trial norm	zscore
Augment training	False, copies=8
Epochs / batch	40 / 16
Early stopping	True, patience=8
LOUO inner-val	user
Random state	42

LOUO results

Variant: BN_GAP — mean=0.9552 ± 0.0584

Held-out user	Inner-val user	n test	n train	Accuracy
11_Mansh	12_Marcus	35	306	1.0000
12_Marcus	14_StephenV2	35	305	1.0000
14_StephenV2	17_Raquel	36	305	0.9722
17_Raquel	18_Bridgette	35	306	1.0000
18_Bridgette	1_Alan	35	306	0.9714
1_Alan	2_Alex	35	306	0.9143
2_Alex	3_Anghad	35	305	0.9714
3_Anghad	4_Daniel	36	305	1.0000
4_Daniel	6_Harry	35	306	0.9429
6_Harry	8_Jestin	35	317	0.9429
8_Jestin	11_Mansh	24	317	0.7917

Final retrain (deployment model)

Variant	Deep
Held-out user	— (trained on all users)
Held-out accuracy	—
Training-set size	376 samples (376 before aug)
Trainable params	204,007
Final train loss	0.0001
Final train acc	1.0000
Model path	saved_models/cnn_Deep_20260512_193155.keras
Labels path	saved_models/labels_Deep_20260512_193155.json
Scaler path	— (per-trial norm in use)